1.) Taxonomic analysis

a. Relative Abundance and Stacked Bar Plot

- Relative abundance was calculated from *species_table.txt*. For each sample, total reads were computed by summing counts across all species. Each species count was then divided by the sample sum and multiplied by 100 to obtain relative abundance in units of percentage.
- The ten most abundant taxa across the whole dataset were identified using *wilkoxmisc::collapse_taxon_table(n = 10)*. This set included **nine named species plus one grouped category ("Minor/Unclassified") that captured low-abundance or unclassified taxa**.
- The top taxa (alphabetical order) were: *Cutibacterium acnes*, *Cutibacterium granulosum*, *Janibacter indicus*, *Klebsiella pneumoniae*, *Kocuria palustris*, *Micrococcus luteus*, Minor/Unclassified, *Moraxella osloensis*, *Pasteurella multocida*, and *Staphylococcus aureus*.
- *Figure 1 A* shows the **stacked bar plot** faceted by Cutotype.
- Cutotype 1 had **more samples** than Cutotype 2 (74 vs 46). Across both Cutotypes, ***Cutibacterium acnes* remained one of the dominant taxa**, indicating a stable core component of the community across groups.

b. Multivariable Association Analysis (MaAsLin3)

- **MaAsLin3** was used to test associations between microbial features and metadata while adjusting for covariates.

- A multivariable model was fitted with City (categorical), Cutotype (categorical), and Age (continuous) as fixed effects. **For simplicity, we did not consider random effects in this model.**
- To specify parameters, we used *TSS* for *normalization*, *LOG* for *transformation*, *0.01* as the *max_significance*, and *standardize = FALSE*, to **remain the unit of Age (year)**.
- MaAsLin3 generated an overview plot summarizing coefficients and significance for both abundance (linear model) and prevalence (logistic model) (*Figure 2 A*).
- **City associations (*Figure 3 A*):**
- The two most significant species (smallest q-values) were *Deinococcus ficus* and *Deinococcus sp. NW-56*.
- City influenced the relative abundance of both taxa. **Dalian exhibited negative associations (coefficients) compared to Baoding, with a greater effect (coefficient) for *Deinococcus sp. NW-56*.**
- **Cutotype associations (*Figure 3 B*):**
- The two most significant species were *Cutibacterium acnes* and *Cutibacterium granulosum*.
- Cutotype influenced the relative abundance of both taxa. **Cutotype 2 showed negative associations (coefficients) relative to Cutotype 1, with a larger effect (coefficient) for *Cutibacterium acnes*,** which was also one of the consistently dominant species across all samples in the **top-10 composition summary** above.

- **Age associations (*Figure 3 C*):**
  - The strongest positive and negative taxa were *Gordonia bronchialis* (positive coefficient) and *Bosea vaviloviae* (negative coefficient).
  - Note that the trendline is from a linear model with *relative_abundance ~ Age*, **the metrics for the trendline are not necessarily the same as the annotated metrics.**
  - **After multiple-testing correction**, both q-values **exceeded 0.01**. This result suggests **no statistically robust age effect on the relative abundance of these species**.

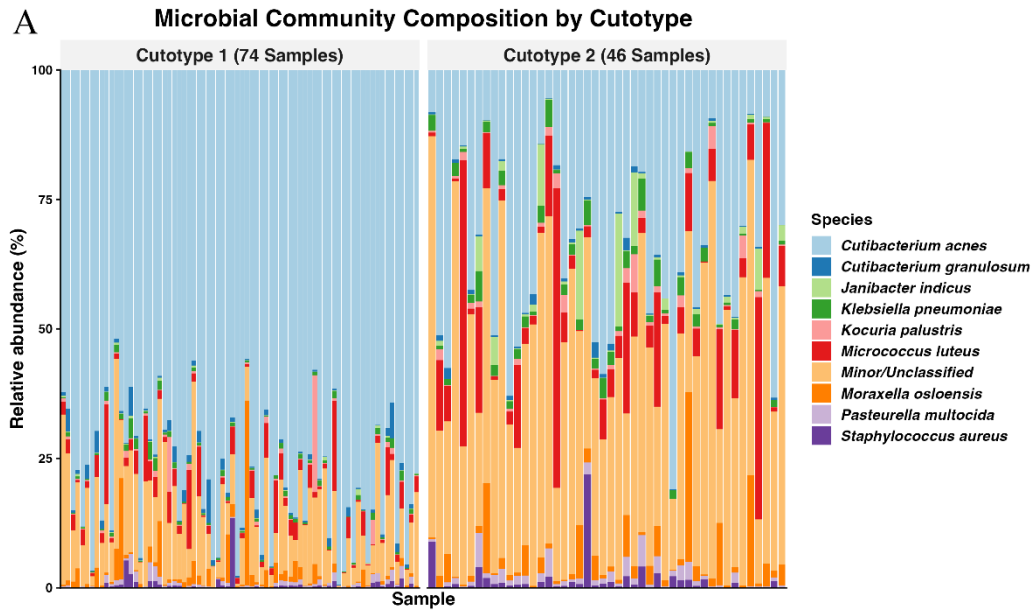


Figure 1. (A) *Stacked bar plot of the ten most abundant species across all samples, faceted by Cutotype.*

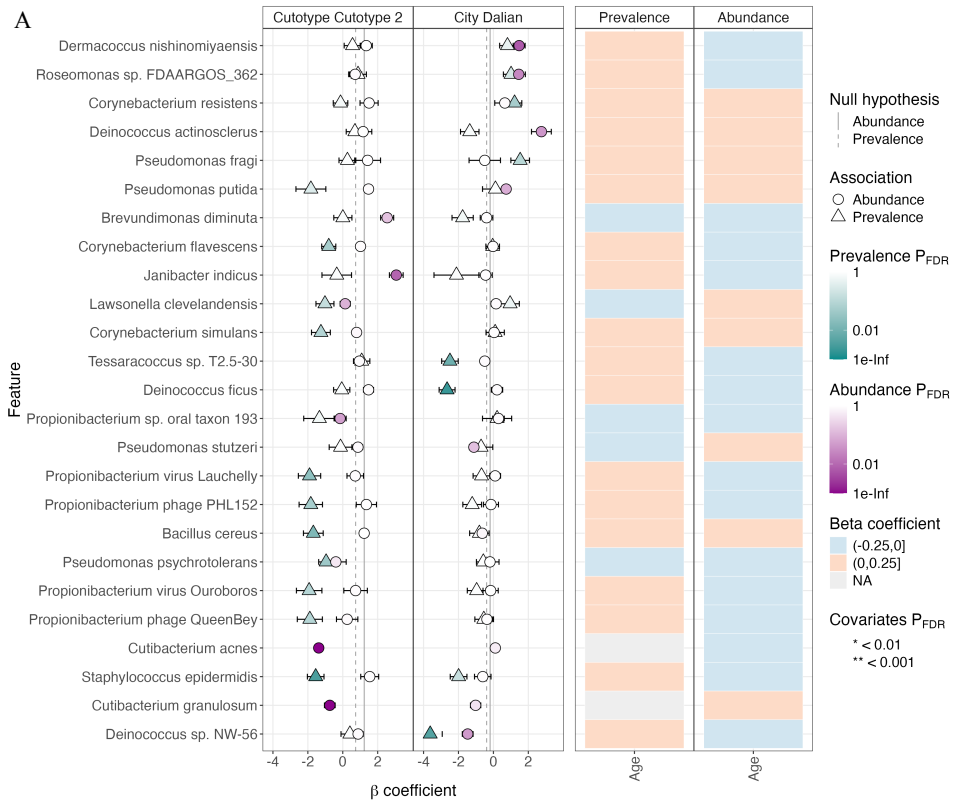


Figure 2. (A) *MaAsLin3 overview plot showing coefficients and statistical significance for both abundance (linear model) and prevalence (logistic model).*

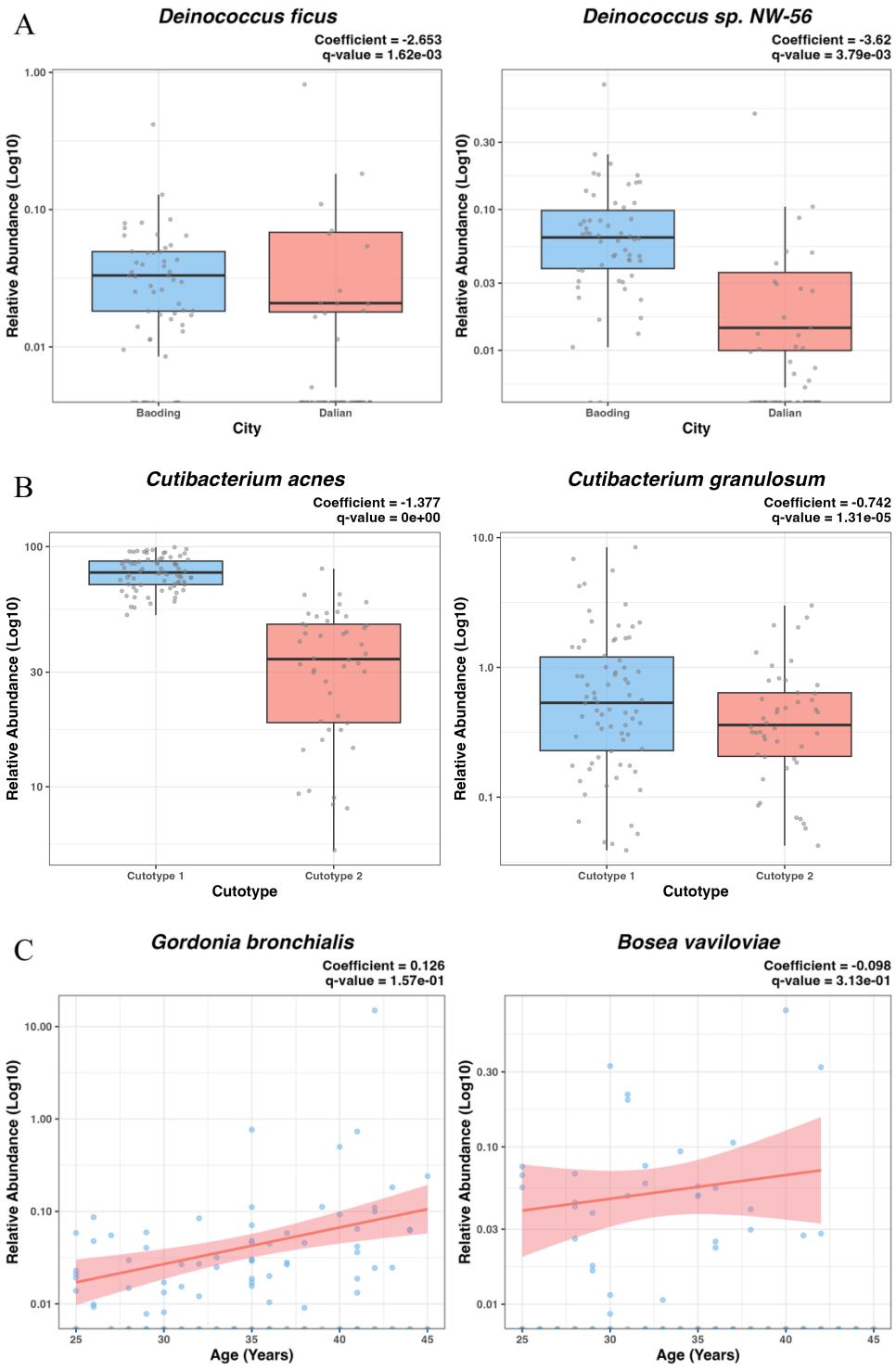


Figure 3. *Selected MaAsLin3 associations. Each panel annotates the association* ***coefficient and q-value****. The y-axis is* ***log-transformed*** *for clarity.* ***(A)*** *City: two most significant taxa.* ***(B)*** *Cutotype: the two most significant taxa.* ***(C)*** *Age: strongest positive and negative taxa; trendlines are shown.*

2.) Correlation Analysis

a. Spearman's Rank Correlation with FDR Correction

- Spearman's rank correlation was computed for **each species–pollutant pair** using *stats::cor.test(method = "spearman")*.
- P-values were adjusted using Benjamini–Hochberg false discovery rate (**BH-FDR**) via *stats::p.adjust(method = "BH")*.
- For each PAH, the most significant associated species (the smallest FDR-adjusted p-value) was summarized in *Table 1* **(For Dibenzo[a,h]anthracene, two species showed the same smallest adjusted p-value).**

b. Scatter Plots for the Strongest Associations

- Five scatter plots were created to visualize the strongest species-PAH associations (smallest q-values from *Table 1*).
- Each plot includes a fitted trendline. Note that the trendline is from a linear model with *relative_abundance ~ PAH*, **the metrics for the trendline are not necessarily the same as the annotated metrics**.
- All five selected associations showed **strong statistical significance** after FDR correction. The **positive correlations (ρ > 0.5)** indicate and support that **higher PAH levels tend to co-occur with higher relative abundance of the corresponding taxa in this dataset**

Species	*PAH*	*ρ*	*S statistic*	*p-value*	*q-value*
Acinetobacter lwoffii	*Fluoranthene*	*0.58*	*121506.69*	*4.73E-12*	*1.46E-07*
Deinococcus sp. NW-56	*Chrysene*	*0.57*	*124577.28*	*1.41E-11*	*2.17E-07*
Deinococcus sp. NW-56	*Benzo[b]fluoranthene*	*0.56*	*127327.48*	*3.61E-11*	*2.79E-07*
Deinococcus sp. NW-56	*Benzo[k]fluoranthene*	*0.56*	*126540.38*	*2.77E-11*	*2.79E-07*
Acinetobacter lwoffii	*Pyrene*	*0.53*	*136725.55*	*7.33E-10*	*2.82E-06*
Acinetobacter lwoffii	*Benzo[a]anthracene*	*0.51*	*141319.16*	*2.85E-09*	*7.99E-06*
Acinetobacter lwoffii	*Benzo[a]pyrene*	*0.48*	*150309.91*	*3.36E-08*	*6.08E-05*
Paracoccus mutanolyticus	*Benzo[ghi]perylene*	*0.47*	*153210.81*	*7.06E-08*	*1.06E-04*
Deinococcus sp. NW-56	*Indeno[1,2,3-cd]pyrene*	*0.44*	*160708.35*	*4.34E-07*	*3.94E-04*
Acinetobacter lwoffii	*Phenanthrene*	*0.36*	*183069.02*	*4.31E-05*	*1.26E-02*
Pectobacterium	*Acenaphtylene*	*0.35*	*187780.39*	*9.86E-05*	*2.32E-02*

carotovorum					
Acinetobacter lwoffii	*Anthracene*	*0.33*	*191932.11*	*1.97E-04*	*3.56E-02*
Mesorhizobium australicum	*Dibenzo[a,h]anthracene*	*0.32*	*194779.54*	*3.11E-04*	*4.55E-02*
Streptacidiphilus sp. DSM 106435	*Dibenzo[a,h]anthracene*	*0.32*	*194779.54*	*3.11E-04*	*4.55E-02*
Massilia sp. NR 4-1	*Fluorene*	*-0.29*	*372449.41*	*1.15E-03*	*1.04E-01*
Pantoea ananatis	*Cotinine*	*0.28*	*206649.20*	*1.78E-03*	*1.32E-01*
Deinococcus wulumuqiensis	*Nicotine*	*-0.28*	*367714.72*	*2.20E-03*	*1.48E-01*
Novosphingobium aromaticivorans	*Acenaphtene*	*-0.28*	*367678.84*	*2.21E-03*	*1.48E-01*

Table 1. *Most significant species association for each PAH (smallest BH-FDR-adjusted p-value). Species, PAH, Spearman's ρ, S statistic, p-value, and FDR-adjusted p-value (q-value) are reported.* ***Ranked by q-value, from small to large. (For Dibenzo[a,h]anthracene, two species showed the equal smallest adjusted p-value.)***

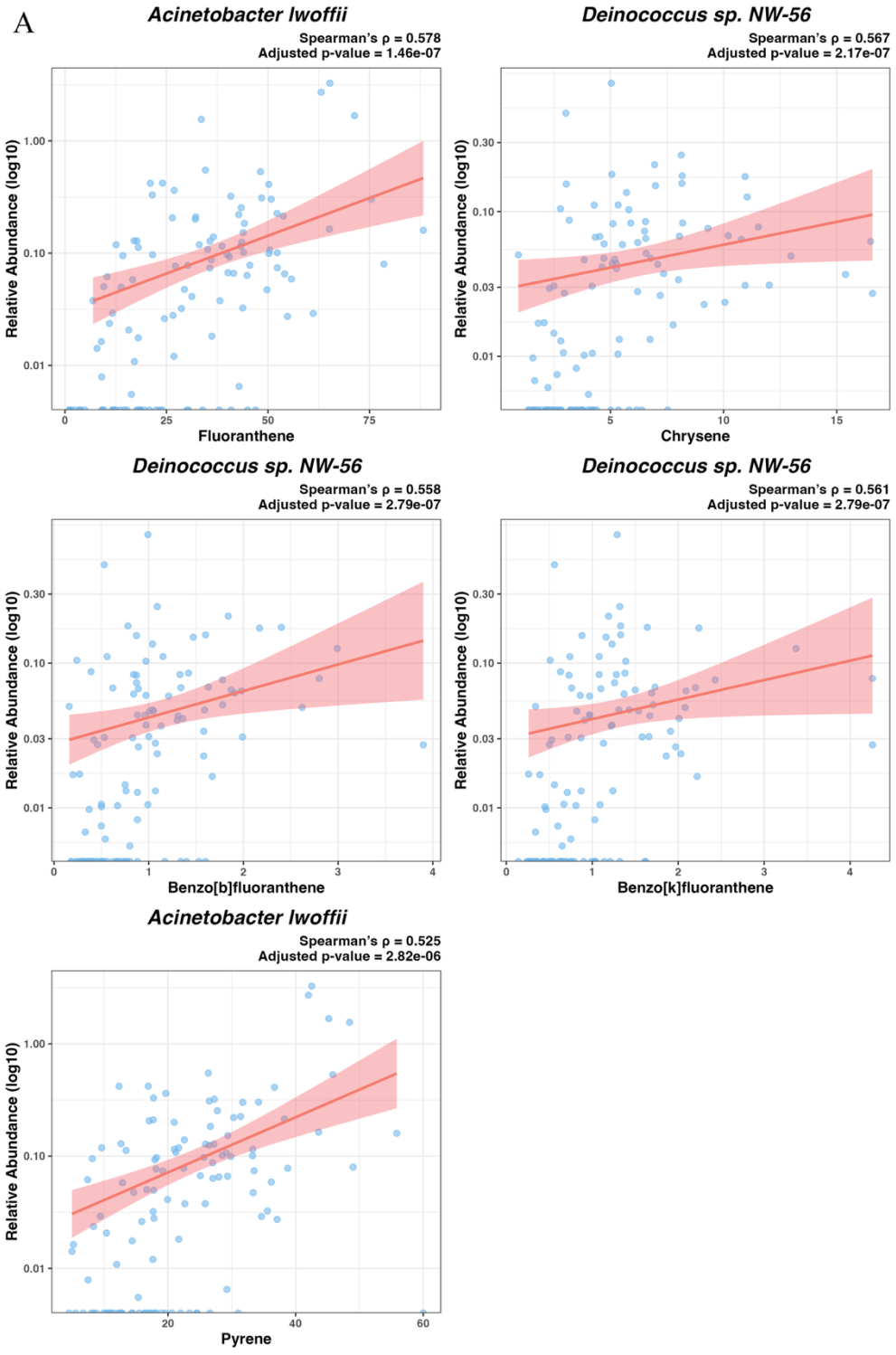


Figure 4. *(A) Scatter plots show the five most significant associations between species relative abundance (log10 scale) and PAH concentration. Each point represents one sample. Annotated statistics report Spearman's ρ and BH–FDR adjusted p-values. Trendlines are added.*

3.) Alpha-diversity analysis

a. Rarefaction and Depth Selection

- ***Importance of rarefaction***
- Rarefaction is a crucial pre-processing strategy for alpha-diversity comparisons, ensuring valid normalization and comparison of abundance or diversity. It makes diversity metrics insensitive to total frequency and depth, which helps reduce significantly fluctuate among samples' information.
- ***Depth selection***
- During rarefaction, a low threshold discards information within each sample. A high threshold discards many samples. The chosen rarefaction depth was set to the **minimum sequencing depth across all samples, retaining all samples while using the highest depth that satisfied this constraint**.
- The selected depth was **27,696 reads** based on these criteria.
- ***Rarefaction steps***
- Rarefaction curves were generated by sampling reads up to each sample's **maximum depth** and computing Shannon diversity at **100 depth points for each sample** using *vegan::rrarefy()* and *vegan::diversity(index = "shannon")*.
- Most curves **approached a plateau near the chosen depth**, supporting the suitability of this cutoff (*Figure 4 A*).

b. Alpha Diversity Analysis

- All samples were rarefied to the chosen depth using *GUniFrac::Rarefy()*.

- The rarefied table was then used to compute **Shannon diversity**, which reflects both richness and evenness, and **species richness**, the number of observed species.
- Density plots (*Figure 5 A*), Q-Q plots (*Figure 5 B*), and violin plots (*Figure 5 C*) were used to inspect metric distributions and compare Cutotypes.
- Density plots (*Figure 5 A*) show **bell-shaped distributions** for both metrics. Q–Q plots (*Figure 5 B*) also show only **small departures from the reference line.** Together, these patterns suggest that both richness and Shannon diversity are **approximately normally distributed in this dataset.**
- Violin plots faceted by Cutotype summarize the full distribution within each group (*Figure 5 C*). They show **higher mean richness and higher mean Shannon diversity in Cutotype 2 than in Cutotype 1. The ANOVA results below provide formal statistical tests for these observed differences.**

c. ANOVA on Alpha Diversity

- For both richness and Shannon diversity, we conducted ANOVA to test the effect of the fixed factors (City, Cutotype, and Age) on alpha diversity.
- ***For richness:***
- First, we check for **assumptions**.
- **Independence:** This is highly related to experimental design; hence, we assume that the observations are independent within each group and that the data are also independent among different groups.

- **Homogeneity of variances for categorical variables:** we conducted a **Levene's Test** using *car::leveneTest()*, the resulting **p-value of 0.1171 > 0.05** illustrates the homogeneity of variances across the **categorical groups**.
- Then we conducted the **ANOVA** using *stats::aov(value ~ City * Cutotype * Age, data = table_richness)* (*Table 2*), followed by a **type III correction** using *car::Anova(type = 3)* to **eliminate the effect of input order** (*Table 3*).
- The results (*Table 3*) show **weak statistical significance** for all variables at a significance level of 0.05.
- We also verified the **normality of residues** of the model: **the large sample size (n = 120)**, the **Shapiro's Test** stats::shapiro.test() with **p-value at 0.67 > 0.05**, and **a Q-Q plot (*Figure 6 A*)** all support the normality of the data.
- Lastly, an **interaction plot** was generated (*Figure 6 C*) to investigate the **specific interactive effects** between the **categorical variables**.
- ***For Shannon:***
- First, we check for **assumptions**.
- **Independence:** This is highly related to experimental design; hence, we assume that the observations are independent within each group and that the data are also independent among different groups.
- **Homogeneity of variances for categorical variables:** we conducted a **Levene's Test** using *car::leveneTest()*, the resulting **p-value of 0.2054 > 0.05** illustrates the homogeneity of variances across the **categorical groups**.

- Then we conducted the **ANOVA** using *stats::aov(value ~ City * Cutotype * Age, data = table_shannon)* (*Table 4*), followed by a **type III correction** using *car::Anova(type = 3)* to **eliminate the effect of input order** (*Table 5*).
- The results (*Table 5*) show that **Cutotypes have a significant main effect on Shannon diversity. Additionally, the interactive effects of (City x Cutotype) and (City x Cutotype x Age) are also statistically significant at a significance level of 0.05.**
- We also verified the **normality of residues** of the model: **the large sample size (n = 120)**, the **Shapiro's Test** stats::shapiro.test() with **p-value at 0.299 > 0.05**, and **a Q-Q plot (*Figure 6 B*)** all support the normality of the data.
- Lastly, an **interaction plot** was generated (*Figure 6 D*) to investigate the **specific interactive effects** between the **categorical variables**. The plot clearly verifies and clarifies the partial results of the ANOVA, showing that **Dalian (City) and Cutotype 1 have a significant effect on the decrease in richness of the microbial community, while the combination of the two categories exacerbates this downward trend.**

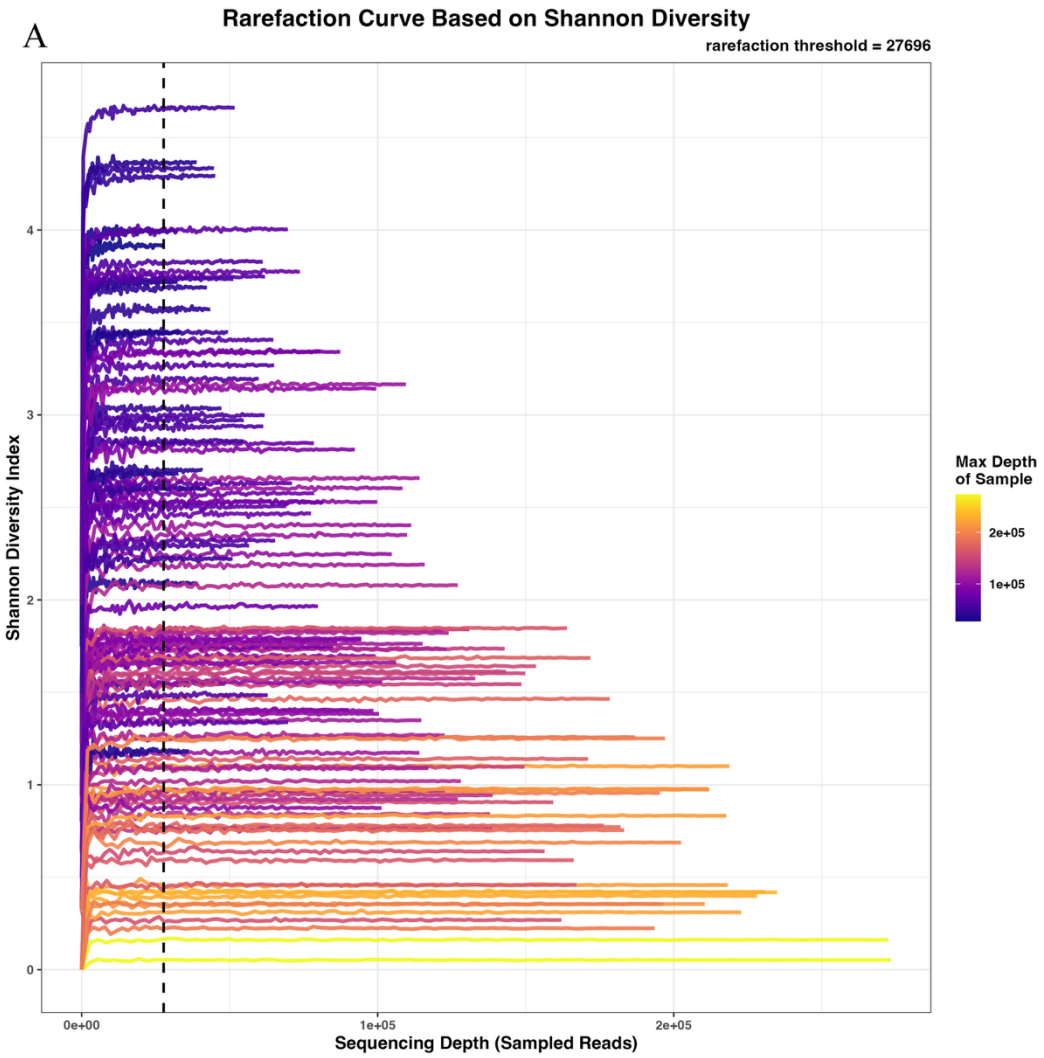


Figure 4 (A) *Rarefaction curves, showing Shannon diversity against sequencing depth for all samples. The selected depth was annotated as a dashed line.*

Tested Term	*F-value*	*p-value*
City	***21.033***	***1.18e-05***
Cutotype	***47.717***	***3.15e-10***
Age	*2.643*	*1.07e-1*
City:Cutotype	***3.930***	***4.99e-2***
City:Age	*0.025*	*8.74e-1*
Cutotype:Age	***5.327***	***2.28e-2***
City:Cutotype:Age	*0.763*	*3.84e-1*

Table 2. ANOVA *results for richness against (City × Cutotype × Age): Tested term, F-statistic, and p-value are listed (Rows with statistical significance are in bold).*

Tested Term	*F-value*	*p-value*
City	*0.5003*	*0.480850*
Cutotype	*0.3918*	*0.532619*
Age	*0.0013*	*0.971016*
City:Cutotype	*0.3060*	*0.581226*
City:Age	*0.0023*	*0.962140*
Cutotype:Age	*1.4967*	*0.223741*
City:Cutotype:Age	*0.7634*	*0.384143*

Table 3. Type III corrected ANOVA *results for richness against (City × Cutotype × Age): Tested term, F-statistic, and p-value are listed (Rows with statistical significance are in bold).*

Tested Term	*F-value*	*p-value*
City	***24.599***	***2.530e-06***
Cutotype	***241.214***	***2.000e-16***
Age	*0.056*	*8.136e-1*
City:Cutotype	***4.634***	***3.35e-2***
City:Age	*1.917*	*1.689e-1*
Cutotype:Age	*1.051*	*3.076e-1*
City:Cutotype:Age	***5.176***	***2.480e-2***

Table 4. ANOVA *results for Shannon diversity against (City × Cutotype × Age): Tested term, F-statistic, and p-value are listed (Rows with statistical significance are in bold).*

Tested Term	*F-value*	*p-value*
City	*0.2737*	*0.60188*
Cutotype	***5.3680***	***0.02233***
Age	*0.0001*	*0.99289*
City:Cutotype	*3.7448*	*0.05549*
City:Age	*0.0014*	*0.97068*
Cutotype:Age	*0.4333*	*0.51172*
City:Cutotype:Age	***5.1755***	***0.02481***

Table 5. Type III corrected ANOVA *results for Shannon diversity against (City × Cutotype × Age): Tested term, F-statistic, and p-value are listed (Rows with statistical significance are in bold).*

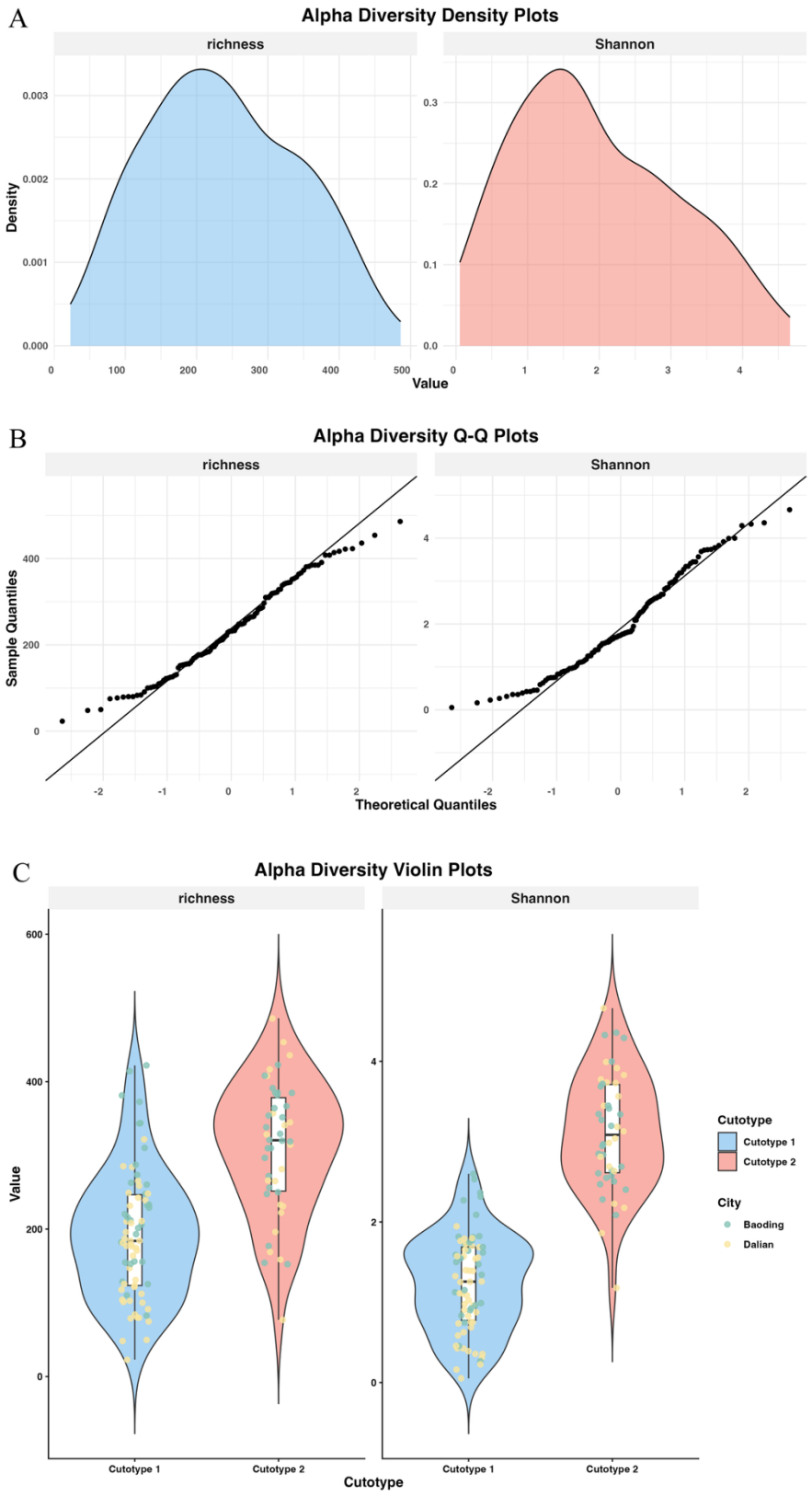


Figure 5. *Alpha-diversity visualization.* ***(A)*** *Density plots for richness and Shannon.* ***(B)*** *Q-Q plots for richness and Shannon.* ***(C)*** *Violin plots for richness and Shannon, faceted by Cutotype.*

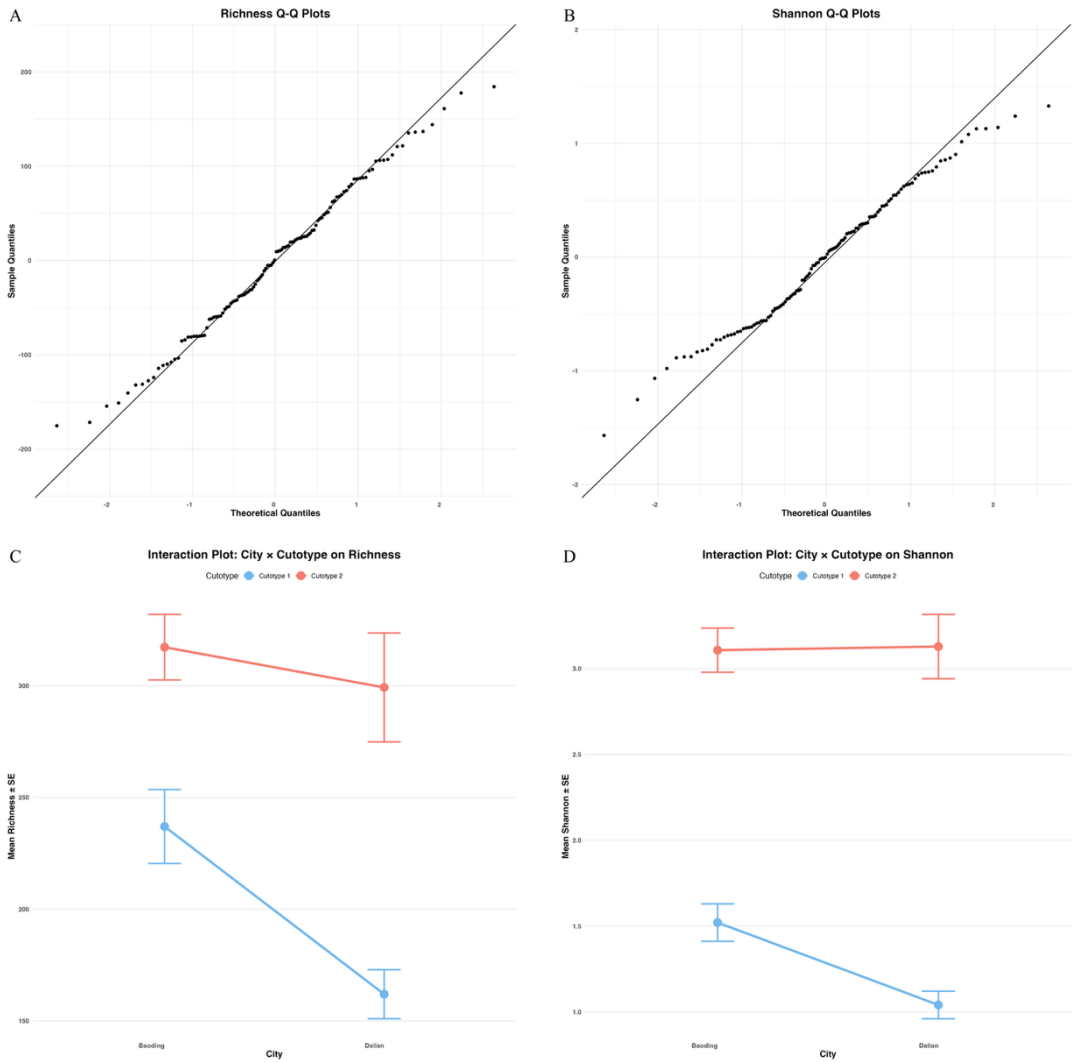


Figure 6. *ANOVA diagnostics and interaction plots.* ***(A)*** *Residual Q-Q plot for richness.* ***(B)*** *Residual Q-Q plot for Shannon diversity.* ***(C)*** *Interaction plot (City × Cutotype) for richness.* ***(D)*** *Interaction plot (City × Cutotype) for Shannon diversity.*

4.) Beta-diversity analysis

a. Distance matrices and PCoA

- Two beta-diversity distance matrices were computed: **Bray–Curtis dissimilarity** with *vegan::vegdist(method = "bray")* and **binary Jaccard distance** with *vegan::vegdist(method = "jaccard", binary = TRUE)*
- For each distance matrix, we performed multidimensional scaling (MDS) – Principal Coordinate Analysis (PCoA) using *stats::cmdscale()*, and manually calculated the **variance explanation percentage** for each axis using the **eigenvalues**.
- *Figure 7* shows two-dimensional ordinations. Points are colored by Cutotype, and shapes indicate City.

b. PERMANOVA

- Permutational Multivariate Analysis of Variance (PERMANOVA) tested whether community composition varied with metadata factors using *vegan::adonis2(by = "margin")*. The model included City, Cutotype, and Age. The *by = "margin"* setting **reduced sensitivity to input ordering**.
- Pseudo-F statistics, R^2 values, and p-values are reported in *Tables 6, 7*.
- For both Bray–Curtis and binary Jaccard distances, **PERMANOVA indicated that City and Cutotype significantly explained variation in community structure ($p < 0.05$), while Age showed no significant effect ($p > 0.05$).**

c. Interpreting Bray–Curtis vs. Jaccard Patterns

- Bray–Curtis and Jaccard **measure different ecological signals**, which explains the different ordination patterns in the PCoA plots.
- **Bray–Curtis** weights taxa by **abundance**, so **dominant and moderately abundant taxa strongly influence distances**.
- **Binary Jaccard** ignores abundance and focuses on whether taxa are **present or absent**. It only reflects differences in **taxonomic membership**.
- This contrast **explained the PCoA ordination results**: Bray–Curtis PCoA showed **clearer separation** between Cutotype 1 and Cutotype 2, with Axis 1 explaining 42% of variance. Binary Jaccard PCoA showed **substantial overlap** between Cutotypes, and Axis 1 explained only 13% of the variance. Together, these patterns suggest **Cutotypes share broadly similar species membership, but differ more in the relative abundance structure of shared taxa.**

d. Biomarkers Usage for Machine Learning

- The Cutotype-associated taxa identified by MaAsLin3 provide **plausible candidate predictors for a supervised classifier**. However, their usage requires **careful consideration.**
- ***Strengths:***

1. Biomarkers selected by MaAsLin3 were **adjusted for multiple testing using FDR control**. This step preserves **strong statistical signals** while **limiting false positives**. The resulting Cutotype-linked features, therefore, rest on a more **reliable statistical basis**.

2. The MaAsLin3 model also tested Cutotype in a **multivariable framework** with several fixed effects. Each reported biomarker exhibits a distinct association with Cutotype after controlling for other covariates. This design **reduces the chance that the signal reflects mixed effects or confounding** from correlated metadata rather than Cutotype itself.
3. Using a defined biomarker set as input features for **reduces the feature dimension**, which **lowers compute and memory costs** during model training and evaluation. It also **improves interpretability**, because model decisions can be traced back to a small set of Cutotype-relevant taxa rather than thousands of weakly informative species.
4. **In this MaAsLin3 analysis**, two Cutotype-linked taxa identified (*Cutibacterium acnes* and *Cutibacterium granulosum*) **appeared among the top ten most abundant species across samples**. *Cutibacterium acnes* was **particularly prevalent and contributed a substantial fraction in many samples**. This pattern supports **robust detection and stable presence of these taxa in the dataset, which strengthens confidence in the observed Cutotype signal**. Additionally, in both the alpha-diversity analysis (Type III ANOVA on Shannon diversity) and the beta-diversity analysis (PERMANOVA), Cutotype is consistently **detected as a significant driver of within-sample and between-sample community differences, either as a main effect or through interaction with other factors**. Together, these results indicate **a strong statistical link between**

Cutotype and microbial community structure. They also support the **feasibility of predicting Cutotype from species-level features.**

- ***Limitations***:
  1. **Unmodeled sources of dependence or confounders can remain**. If hidden batch effects or participant-level structure exist, effect estimates can shift, and selected biomarkers may not replicate. Thus, random effects should be considered carefully and processed during MaAsLin3.
  2. MaAsLin3 focuses on and tests each single dependent variable's effect on the independent variable. However, a classifier may rely on **feature interactions and joint effects terms**. Using these biomarkers directly may not maximize predictive accuracy. As a result, testing for interactive effects and evaluation and modeling, such as ANOVA, are recommended.
  3. Biomarker discovery and model training on the same dataset have possibilities for **information leakage if not processed properly**, which leads to overestimation of model performance and overfitting. A correct workflow must isolate feature between training and testing dataset, such as k-fold cross-validation, and leave-one-data set out (LODO).
- Therefore, while the identified biomarkers are **suitable as candidate features for ML models**, their usage should be **carefully considered and processed.**

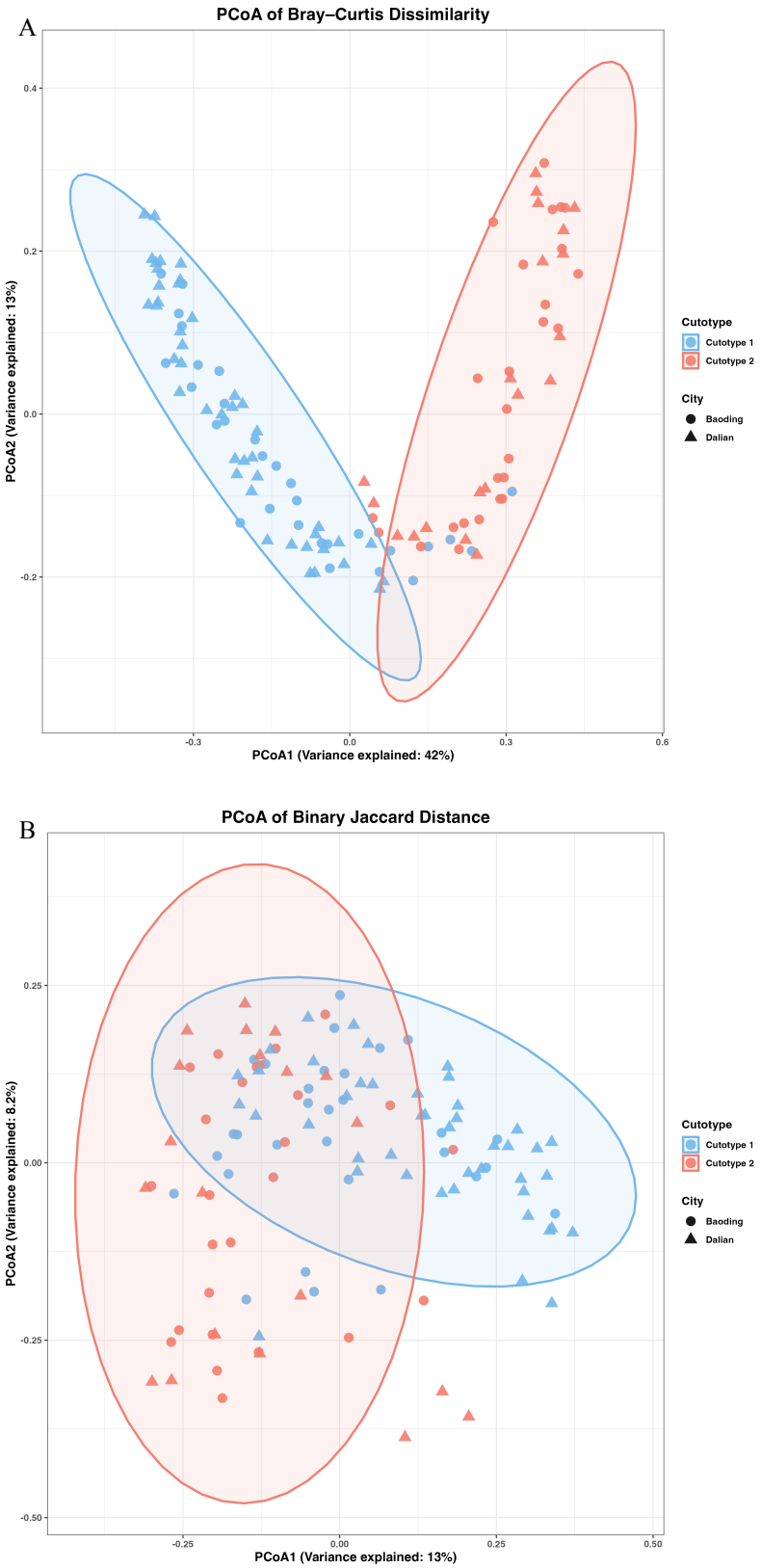


Figure 7. *PCoA ordinations, colored by Cutotype and shaped by City. Ellipses show 95% confidence regions.* ***(A)*** *Bray–Curtis distance.* ***(B)*** *Binary Jaccard distance.*

Tested term	*F statistic*	*R²*	*p-value*
City	***2.4656***	***0.01439***	***0.038***
Cutotype	***48.1571***	***0.28108***	***0.001***
Age	*0.9874*	*0.00576*	*0.367*

Table 6. *PERMANOVA results for Bray–Curtis distance, showing pseudo-F, R², and p-value (Rows with statistical significance are in bold).*

Tested term	*F statistic*	*R²*	*p-value*
City	***2.9230***	***0.02312***	***0.001***
Cutotype	***5.3994***	***0.04271***	***0.001***
Age	*1.3791*	*0.01091*	*0.058*

Table 7. *PERMANOVA results for binary Jaccard distance, showing pseudo-F, R², and p-value (Rows with statistical significance are in bold).*

Codes

1.) Taxonomic analysis

```r
#################################################################
# Preparation
#################################################################

# Load packages (Install them first if have not)
library(tidyverse)
library(plyr)
library(wilkoxmisc)
library(maaslin3)
library(patchwork)

# Set seed for random processes
set.seed(1)

#################################################################
# 1.a. Relative Abundance & Stacked Bar Plot
#################################################################

# Read the table into R
OTU <- read_tsv("species_table.txt")
names(OTU)[1] <- c("Sample")

# Calculate the count of each taxon in each sample and only remove taxa not identified
OTU <- OTU %>%
  gather(key = "Species", value = "Count", -Sample) %>%
  filter(Count > 0)

# Tabulate read counts by species
OTU <- ddply(OTU, .(Sample, Species), summarise, count = sum(Count))

# Convert count to relative abundance and add column (require dplyr)
# RelativeAbundance in unit %
OTU <- ddply(OTU, .(Sample), mutate, RelativeAbundance = (count * 100) / sum(count))

# Save the processed table into a tsv file
write_tsv(OTU,"OTU_count_adundance_long.tsv")
```

```r

# Prepare the table for plotting

# Read the previously saved tsv file
OTU <- read_tsv("OTU_count_adundance_long.tsv")

# Collapse taxa table to only 10 top phyla, genus, family, etc.
if ("package:plyr" %in% search()) {
  detach("package:plyr", unload = TRUE)
}
OTUTable <- collapse_taxon_table(OTU, n = 10, Rank = "Species")

# Read the metadata table
Meta <- read_tsv("metadata.txt")

# Merge relative abundance table and metatable together
OTUTable <- merge(OTUTable, Meta, by = "Sample", all.x = TRUE)

# Save the table in tsv file
write_tsv(OTUTable, "Top10Species.tsv")

# Plotting
OTUTable <- read_tsv("Top10Species.tsv")

# Set factor for two variables
OTUTable$Species <- factor(OTUTable$Species, levels = unique(OTUTable$Species))
OTUTable$Sample <- factor(OTUTable$Sample,levels = unique(OTUTable$Sample))

# Add sample count to Cutotype label (for polt)
OTUTable <- OTUTable %>%
  group_by(Cutotype) %>%
  mutate(Cutotype_Label = paste0(Cutotype, " (", n_distinct(Sample), " Samples)")) %>%
  ungroup()

labels <- unique(OTUTable$Cutotype_Label)
OTUTable$Cutotype_Label <- factor(OTUTable$Cutotype_Label, levels = sort(labels))

# Plot the relative abundance
```

```r
(plot_stacked_bar_chart <- ggplot(OTUTable, aes(x = Sample, y = RelativeAbundance, fill = Species)) +
    geom_bar(stat="identity") +
    scale_fill_brewer(palette = "Paired") +
    facet_wrap(~Cutotype_Label, scales = "free_x") +
    scale_y_continuous(expand = c(0, 0)) +
    ylab("Relative abundance (%)") +
    xlab("Sample") +
    labs(title = "Microbial Community Composition by Cutotype") +
    theme_classic() +
    theme(
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      plot.margin = margin(15, 15, 15, 15),
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank(),
      legend.position = "right",
      legend.text = element_text(face = "bold.italic", size = 10),
      legend.title = element_text(face = "bold"),
      legend.key.size = unit(0.5, "cm"),
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12)
    ))

# Save the plot in pdf file
ggsave("fig_1a_composition_stacked_bar.png", plot_stacked_bar_chart, width = 10, height = 6, dpi = 300)

##################################################################
# 1.b. Multivariable Association Analysis
##################################################################

# Data Preparation

# Input taxa table
OTU_abundance_long <- read_tsv("OTU_count_adundance_long.tsv")
OTU_abundance_wide <- OTU_abundance_long %>%
  select(Sample, Species, RelativeAbundance) %>%
  pivot_wider(
```

```r
    names_from = Species,
    values_from = RelativeAbundance,
    values_fill = 0
  )
input_data <- OTU_abundance_wide
input_data <- as.data.frame(input_data)
names(input_data)[1] <- c("Sample")
write_tsv(input_data,"OTU_adundance_wide.tsv")
input_data <- read_tsv("OTU_adundance_wide.tsv") %>%
as.data.frame()
rownames(input_data) <- input_data$Sample
input_data$Sample <- NULL

# Input meta data
input_metadata <- read_tsv("metadata.txt")
input_metadata <- as.data.frame(input_metadata)
rownames(input_metadata) <- input_metadata$Sample
input_metadata$Sample <- NULL
input_metadata <- input_metadata %>%
  mutate(
    City     = factor(City, levels =
unique(input_metadata$City)),
    Cutotype = factor(Cutotype, levels =
unique(input_metadata$Cutotype)),
    Age      = as.numeric(Age)
  )

# Output directory
output_dir <- "1b_maaslin3_output"

# Conduct MaAsLin3 analysis
fit_out <- maaslin3 (
  input_data = input_data,
  input_metadata = input_metadata,
  output = output_dir,
  formula = '~ City + Cutotype + Age',
  normalization = 'TSS',
  transform = 'LOG',
  augment = TRUE,
  standardize = FALSE, # remain interpretability for Age in
unit of year
  max_significance = 0.01,
  median_comparison_abundance = TRUE,
  median_comparison_prevalence = TRUE,
```

```r
  max_pngs = 250,
  cores = 1
  )

# Interpretation and visualization
res <- read_tsv(paste0(output_dir, "/all_results.tsv"))

# A function to search for target species
get_top_species <- function(res_table, variable_name, direction = "both", num = 2) {
  subset_df <- res_table %>%
    filter(metadata == variable_name) %>%
    arrange(qval_individual)

  if(direction == "both") {
    return(subset_df %>% head(num))
  } else if (direction == "positive") {
    return(subset_df %>% filter(coef > 0) %>% head(num))
  } else if (direction == "negative") {
    return(subset_df %>% filter(coef < 0) %>% head(num))
  }
}

# City (Categorical)
top_city <- get_top_species(res, "City", "both", 2)
(top_city_species <- top_city$feature)

# Cutotype (Categorical)
top_cutotype <- get_top_species(res, "Cutotype", "both", 2)
(top_cutotype_species <- top_cutotype$feature)

# C. Age (Continuous)
top_age_pos <- get_top_species(res, "Age", "positive", 1)
(top_age_pos_species <- top_age_pos$feature)
top_age_neg <- get_top_species(res, "Age", "negative", 1)
(top_age_neg_species <- top_age_neg$feature)

# Visualization

# Box plot for categorical variables

# A function for subplots
plot_boxplot <- function(species_name, meta_col, res_row) {
  # Extract relative abundance & metadata
```

```r
  plot_data <- data.frame(
    Abundance = input_data[[species_name]],
    Group = input_metadata[[meta_col]]
  )

  # Annotate coefficient and q-value
  stats_text <- paste0("Coefficient = ", round(res_row$coef, 3),
                       "\nq-value = ", format(res_row$qval_individual, scientific = TRUE, digits = 3))

  # Plot (subplot)
  ggplot(plot_data, aes(x = Group, y = Abundance, fill = Group)) +
    geom_boxplot(alpha = 0.7, outlier.shape = NA) +
    geom_jitter(width = 0.2, size = 1, alpha = 0.5, color = "grey50") +
    scale_y_log10() + # for better appearance, automatically remove 0 relative abundance
    scale_fill_manual(values = c("#71b7ed", "#f57c6e")) +
    labs(
      title = species_name,
      subtitle = stats_text,
      x = meta_col,
      y = "Relative Abundance (Log10)",
      ) +
    theme_bw() +
    theme(
      plot.title = element_text(face = "bold.italic", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      legend.position = "none",
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
      )
}

# Plot subplots
(p1 <- plot_boxplot(top_city_species[1], "City", top_city[1,]))
```

```r
(p2 <- plot_boxplot(top_city_species[2], "City", top_city[2,]))
(p3 <- plot_boxplot(top_cutotype_species[1], "Cutotype", top_cutotype[1,]))
(p4 <- plot_boxplot(top_cutotype_species[2], "Cutotype", top_cutotype[2,]))

# Merge and save subplots
(p_categorical_city <- (p1 + p2))
(p_categorical_cutotype <- (p3 + p4))
ggsave("fig_1b_city_box_maaslin3.png", p_categorical_city, width = 10, height = 5, dpi = 300)
ggsave("fig_1b_cutotype_box_maaslin3.png", p_categorical_cutotype, width = 10, height = 5, dpi = 300)


# Scatter plots for continuous variable

# A function for subplots
plot_scatter <- function(species_name, res_row) {
  # Extract relative abundance & metadata
  plot_data <- data.frame(
    Abundance = input_data[[species_name]],
    Age = input_metadata[["Age"]]
  )

  # Annotate coefficient and q-value
  stats_text <- paste0("Coefficient = ", round(res_row$coef, 3),
                       "\nq-value = ", format(res_row$qval_individual, scientific = TRUE, digits = 3))

  # Plot (subplot)
  ggplot(plot_data, aes(x = Age, y = Abundance)) +
    geom_point(alpha = 0.6, color = "#71b7ed") +
    geom_smooth(method = "lm", color = "#f57c6e", fill = "#f2656f") +
    scale_y_log10() +
    labs(title = species_name,
         subtitle = stats_text,
         x = "Age (Years)",
         y = "Relative Abundance (Log10)") +
    theme_bw() +
```

```
    theme(
      plot.title = element_text(face = "bold.italic", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      legend.position = "none",
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
      )
}

# Plot subplots
(p5 <- plot_scatter(top_age_pos_species, top_age_pos))
(p6 <- plot_scatter(top_age_neg_species, top_age_neg))

# Merge and save subplots
(p_continuous <- p5 + p6)
ggsave("figure_1b_age_scatter_maaslin3.png", p_continuous, width = 10, height = 5, dpi = 300)
```

2.) Correlation Analysis

```r
#################################################################
# Preparation
#################################################################

# Load packages (Install them first if have not)
library(tidyverse)
library(patchwork)
library(dplyr)
library(tidyr)
library(readr)

# Set seed for random processes
set.seed(1)

#################################################################
# 2.a. Spearman + FDR
#################################################################

# Read the table into R
taxa_data <- read_tsv("OTU_adundance_wide.tsv") %>%
as.data.frame()
rownames(taxa_data) <- taxa_data$Sample
taxa_data$Sample <- NULL

pollutant_data <- read_tsv("pollutant.txt") %>%
as.data.frame()
rownames(pollutant_data) <- pollutant_data$Sample
pollutant_data$Sample <- NULL

# Multiple Spearman's rank-order correlation analysis (use a
nested loop)
species_names <- colnames(taxa_data)
pah_names <- colnames(pollutant_data)
cor_list <- list()

for (pah in pah_names) {
  for (sp in species_names) {

    x <- as.numeric(pollutant_data[[pah]])
    y <- as.numeric(taxa_data[[sp]])

    ct <- cor.test(x, y, method = "spearman")
```

```r

      cor_list[[length(cor_list) + 1]] <- data.frame(
        Species = sp,
        PAH = pah,
        rho = unname(ct$estimate),
        S_score = unname(ct$statistic),
        p_value = ct$p.value,
        n = length(x)
      )
    }
  }

cor_results <- bind_rows(cor_list)

# FDR correction (Benjamini-Hochberg method)
cor_results <- cor_results %>%
  mutate(adj_q_value = p.adjust(p_value, method = "BH"))

# Write all results
write_tsv(cor_results, "spearman_all.tsv")

# find most significant species for each PAHs
cor_significant <- cor_results %>%
  group_by(PAH) %>%
  slice_min(adj_q_value, n = 1, with_ties = TRUE) %>%
  ungroup() %>%
  arrange(adj_q_value)
write_tsv(cor_significant, "spearman_significant.tsv")

#################################################################
# 2.b. Spearman + FDR
#################################################################

# Prepare data
cor_plot <- read_tsv("spearman_significant.tsv") %>%
  arrange(adj_q_value) %>%
  head(5)

# A function for subplots
plot_scatter <- function(species, pah, rho_val, adj_q_val) {
  # Extract relative abundance & PAH
  df <- data.frame(
    RA = as.numeric(taxa_data[[species]]),
    PAH = pollutant_data[[pah]]
```

```r
  )

  # Annotate rho and adjusted q-value
  stats_text <- paste0(
    "Spearman's ρ = ", round(rho_val, 3),
    "\nAdjusted p-value = ", format(adj_q_val, digits = 3, scientific = TRUE)
  )

  # Plot (subplot)
  ggplot(df, aes(x = PAH, y = RA)) +
    geom_point(color = "#71b7ed", alpha = 0.6, size = 2) +
    geom_smooth(method = "lm", color = "#f57c6e", fill = "#f2656f") +
    scale_y_log10() +
    labs(
      title = species,
      subtitle = stats_text,
      x = pah,
      y = "Relative Abundance (log10)"
    ) +
    theme_bw() +
    theme(
      plot.title = element_text(face = "bold.italic", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      legend.position = "none",
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
    )
}

# Plot subplots
plot_list <- list()
for (i in 1:nrow(cor_plot)) {
  row <- cor_plot[i, ]
  p <- plot_scatter(
    species = row$Species,
    pah = row$PAH,
    rho_val = row$rho,
    adj_q_val = row$adj_q_value
```

```
  )
  plot_list[[i]] <- p
}

# Merge and save subplots
(final_plot <- wrap_plots(plotlist = plot_list, ncol = 2))
ggsave("fig_2b_cor_scatter.png", final_plot, width = 10,
height = 15, dpi = 300)
```

3.) Alpha-diversity analysis

```
############################################################
# Preparation
############################################################

# Load packages (Install them first if have not)
library(readr)
library(dplyr)
library(tidyr)
library(ggplot2)
library(vegan)
library(OTUtable)
library(car)
library(viridis)
library(reshape2)
library(GUniFrac)

# Set seed for random processes
set.seed(1)

############################################################
# 3.a. Rarefaction
############################################################

# Generate long form data
OTU_long <- read_tsv("OTU_count_adundance_long.tsv")
OTU_long <- OTU_long[, c(1:ncol(OTU_long)-1)]
write_tsv(OTU_long,"OTU_count_long.tsv")
OTU_long <- read_tsv("OTU_count_long.tsv")

# Generate wide form data
OTU_wide <- OTU_long %>%
  pivot_wider(
    names_from = Species,
    values_from = count,
    values_fill = 0
  )
colnames(OTU_wide) <- c("Sample", paste0("Species",
1:ncol(OTU_wide)))
write_tsv(OTU_wide,"OTU_count_wide.tsv")
OTU_wide <- read_tsv("OTU_count_wide.tsv") %>%
as.data.frame()
rownames(OTU_wide) <- OTU_wide$Sample
```

```r
OTU_wide$Sample <- NULL

# A function for calculate shannon for each depth in each sample
compute_rarefy_shannon <- function(count_sample) {
  # Depth and points in each sample
  max_depth <- sum(count_sample)
  depths <- seq(10, max_depth, length.out = 100)

  # Sample and calculate shannon for different depth
  shannon_values <- sapply(depths, function(d) {
    rarefied <- rrarefy(count_sample, sample = d)
    diversity(rarefied, index = "shannon")
  })

  # Export
  data.frame(depth = depths, shannon = shannon_values)
}

# Calculate shannon for each depth in each sample
rare_list <- lapply(rownames(OTU_wide), function(s) {
  df <- compute_rarefy_shannon(OTU_wide[s, ])
  df$Sample <- s
  df
})
rare_df <- bind_rows(rare_list)
write_tsv(rare_df, "shannon_depth_rarefaction.tsv")
rare_df <- read_tsv("shannon_depth_rarefaction.tsv")

# Choose rarefaction depth
(chosen_depth <- min(rowSums(OTU_wide)))

# Add `max_depth` col in data (for coloring)
sample_max_depth <- tapply(rare_df$depth, rare_df$Sample, max)
rare_df$max_depth <- sample_max_depth[ rare_df$Sample ]

# Plot & save
(plot_rare <- ggplot(rare_df, aes(x = depth, y = shannon, group = Sample)) +
  geom_line(aes(color = max_depth), size = 1.2, alpha = 0.9) +
  geom_vline(xintercept = chosen_depth, linetype = "dashed", size = 0.8, color = "black") +
```

```r
  scale_color_viridis(option = "plasma", direction = 1) +
  theme_bw() +
  labs(
    title = "Rarefaction Curve Based on Shannon Diversity",
    x = "Sequencing Depth (Sampled Reads)",
    y = "Shannon Diversity Index",
    color = "Max Depth \nof Sample",
    subtitle = paste0("rarefaction threshold = ", chosen_depth)
  ) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold"),
    legend.text = element_text(face = "bold"),
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.title = element_text(face = "bold", size = 12),
    axis.text = element_text(face = "bold"),
    plot.subtitle = element_text(face = "bold", hjust = 1)
  ))

ggsave("fig_3a_rarefaction_curve.png", plot_rare, width = 10, height = 10, dpi = 300)

# Generate rarefied tables
rarefied_table <- Rarefy(OTU_wide, depth = chosen_depth)$otu.tab.rff
rarefied_table_long <- as.data.frame(as.table(rarefied_table))
names(rarefied_table_long)[1:3] <- c("Sample", "Species", "Count")
rarefied_table_long <- rarefied_table_long %>%
  filter(! Count == 0)
write_tsv(rarefied_table_long,"rarefied_table_long.tsv")
rarefied_table_wide <- dcast(rarefied_table_long, Species ~ Sample, value.var = "Count", fill = 0)
write_tsv(rarefied_table_wide,"rarefied_table_wide.tsv")

############################################################
# 3.b. Rarefaction
############################################################

# Calculate richness
richness <- read_tsv("rarefied_table_long.tsv") %>%
```

```r
  select(Sample) %>%
  mutate(richness = 1) %>%
  group_by(Sample) %>%
  mutate(richness= sum(richness)) %>%
  ungroup() %>%
  unique()
write_tsv(richness,"alpha_diversity_richness.tsv")

# Caculate shannon diversity、
# Transpose the data frame
alpha <- read_tsv("rarefied_table_wide.tsv")
Sample <- colnames(alpha[2:ncol(alpha)])
Species <- alpha[1:nrow(alpha), 1]
alpha[,1] <- NULL
colnames(alpha) <- NULL
alpha_t <- data.frame(t(alpha))
alpha_t[is.na(alpha_t)] <- 0
colnames(alpha_t) <- t(Species)
# Calculate the shannon diversity for global samples
Shannon <- diversity(alpha_t, "shannon")
# Merge the sample info with alpha diversity
data <- data.frame(cbind(Sample, Shannon))
# Write out
write_tsv(data, "alpha_diversity_shannon.tsv")

# Merge alpha diversity results
richness <- read_tsv("alpha_diversity_richness.tsv")
shannon <- read_tsv("alpha_diversity_shannon.tsv")
merge <- left_join(richness, shannon)
write_tsv(merge,"alpha_diversity_merge.tsv")

# Visualization

# Read the alpha diversity table
table <- read_tsv("alpha_diversity_merge.tsv")

# Reshape the data to long format and filter out zero
table <- table %>%
  pivot_longer(cols = 2:ncol(table), names_to = "alpha", values_to = "value") %>%
  filter(value > 0)

# Read metadata and merge with the alpha diversity data
```

```r
meta <- read_tsv("metadata.txt")
table <- left_join(table, meta, by = "Sample")

# Density plot
(density_plot <- ggplot(table, aes(x = value, fill = alpha)) +
    geom_density(alpha = 0.5) +
    facet_wrap(~ alpha, scales = "free") +
    scale_fill_manual(values = c("#71b7ed", "#f57c6e")) +
    labs(
     title = "Alpha Diversity Density Plots",
     x = "Value",
     y = "Density"
     ) +
    theme_minimal() +
    theme(
     plot.margin = margin(15, 15, 15, 15),
     plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
     axis.title = element_text(face = "bold", size = 12),
     axis.text = element_text(face = "bold"),
     axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
     legend.position = "none",
     strip.background = element_rect(fill = "grey95", color = NA),
     strip.text = element_text(face = "bold", size = 12),
     plot.subtitle = element_text(face = "bold", hjust = 1)
     ))
ggsave("fig_3b_alpha_diversity_density_plots.png", plot = density_plot, width = 10, height = 5, dpi = 300)

# QQ plot
(qq_plot <- ggplot(table, aes(sample = value)) +
   stat_qq() +
   stat_qq_line() +
   facet_wrap(~ alpha, scales = "free") +
   theme_minimal() +
   labs(
     title = "Alpha Diversity Q-Q Plots",
     x = "Theoretical Quantiles",
     y = "Sample Quantiles"
   ) +
   theme_minimal() +
```

```r
  theme(
    plot.margin = margin(15, 15, 15, 15),
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.title = element_text(face = "bold", size = 12),
    axis.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
    legend.position = "none",
    strip.background = element_rect(fill = "grey95", color = NA),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(face = "bold", hjust = 1)
  ))
ggsave("fig_3b_alpha_diversity_qq_plots.png", plot = qq_plot, width = 10, height = 5, dpi = 300)

# Violin plot
(violin_plot <- ggplot(table, aes(Cutotype, value)) +
   geom_violin(aes(fill = Cutotype), trim = FALSE, alpha = 0.6) +
   geom_boxplot(width = 0.1, outlier.shape = NA) +
   geom_jitter(aes(color = City), width = 0.1, size = 2, alpha = 0.8) +
   theme_classic() +
   facet_wrap(~alpha, scales = "free") +
   scale_fill_manual(values = c(
     "Cutotype 1" = "#71b7ed",
     "Cutotype 2" = "#f57c6e"
   )) +
   scale_color_manual(values = c(
     "Baoding" = "#84c3b7",
     "Dalian" = "#fae69e"
   )) +
   labs(
     title = "Alpha Diversity Violin Plots",
     x = "Cutotype",
     y = "Value"
     ) +
   theme(
     plot.margin = margin(15, 15, 15, 15),
     plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
     axis.title = element_text(face = "bold", size = 12),
```

```r
      axis.text = element_text(face = "bold"),
      axis.text.x = element_text(angle = 0, vjust = 0.5),
      legend.position = "right",
      legend.title = element_text(face = "bold"),
      legend.text = element_text(face = "bold"),
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
      ))
ggsave("fig_3b_alpha_diversity_violin_plots.png", violin_plot, width = 10, height = 8, dpi = 300)

############################################################
# 3.c. ANOVA
############################################################

# ANOVA for simultaneous effect analysis

# Data
table_richness <- subset(table, alpha == "richness")
table_shannon <- subset(table, alpha == "Shannon")

# For richness

# Assumptions
# Independence: Independence is highly related to experimental design, hence we assume the observations are independent within the single group, and data are also independent among different groups.
# Normality: need residues from the model, hence we verify this later
# Homogeneity of variances for categorical variables (Levene's Test):
leveneTest(value ~ City * Cutotype, data = table_richness, center = median)

# ANOVA
aov_richness <- aov(value ~ City * Cutotype * Age, data = table_richness)
(aov_richness_typeIII <- Anova(aov_richness, type = 3))

# Verify normality
# Shapiro-Wilk Test
```

```r
shapiro.test(residuals(aov_richness))
# QQ plot
df_resid_richness <- data.frame(residuals = residuals(aov_richness))
(qq_plot_richness <- ggplot(df_resid_richness, aes(sample = residuals)) +
    stat_qq() +
    stat_qq_line() +
    theme_minimal() +
    labs(
      title = "Richness Q-Q Plots",
      x = "Theoretical Quantiles",
      y = "Sample Quantiles"
    ) +
    theme_minimal() +
    theme(
      plot.margin = margin(15, 15, 15, 15),
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
      legend.position = "none",
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
    ))
ggsave("fig_3c_richness_qq_plots.png", plot = qq_plot_richness, width = 10, height = 10, dpi = 300)

# Interaction Plot
plot_data_richness <- table_richness %>%
  group_by(City, Cutotype) %>%
  summarise(
    mean_value = mean(value),
    se_value = sd(value) / sqrt(n()),
    .groups = "drop"
  )
(interaction_plot_richness <- ggplot(
  plot_data_richness,
  aes(x = City, y = mean_value, group = Cutotype, color = Cutotype)
```

```r
) +
  geom_point(size = 5) +
  geom_line(linewidth = 1.5) +
  geom_errorbar(
    aes(ymin = mean_value - se_value, ymax = mean_value + se_value),
    width = 0.15,
    linewidth = 1
  ) +
  scale_color_manual(values = c("#71b7ed", "#f57c6e")) +
  labs(
    title = "Interaction Plot: City × Cutotype on Richness",
    x = "City",
    y = "Mean Richness ± SE",
    color = "Cutotype"
  ) +
  theme_minimal() +
  theme(
    panel.grid.major.y = element_line(color = "grey95", linewidth = 1),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    plot.margin = margin(15, 15, 15, 15),
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.title = element_text(face = "bold", size = 12),
    axis.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
    legend.position = "top",
    strip.background = element_rect(fill = "grey95", color = NA),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(face = "bold", hjust = 1)
  ))
ggsave("fig_3c_richness_interaction_plot.png", interaction_plot_richness, width = 10, height = 10, dpi = 300)

# For shannon

# Assumptions
```

```r
# Independence: Independence is highly related to experimental design, hence we assume the observations are independent within the single group, and data are also independent among different groups.
# Normality: need residues from the model, hence we verify this later
# Homogeneity of variances for categorical variables (Levene's Test):
leveneTest(value ~ City * Cutotype, data = table_shannon, center = median)

# ANOVA
aov_shannon <- aov(value ~ City * Cutotype * Age, data = table_shannon)
(aov_shannon_typeIII <- Anova(aov_shannon, type = 3))

# Verify normality
# Shapiro-Wilk Test
shapiro.test(residuals(aov_shannon))
# QQ plot
df_resid_shannon <- data.frame(residuals = residuals(aov_shannon))
(qq_plot_shannon <- ggplot(df_resid_shannon, aes(sample = residuals)) +
    stat_qq() +
    stat_qq_line() +
    theme_minimal() +
    labs(
      title = "Shannon Q-Q Plots",
      x = "Theoretical Quantiles",
      y = "Sample Quantiles"
    ) +
    theme_minimal() +
    theme(
      plot.margin = margin(15, 15, 15, 15),
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
      legend.position = "none",
      strip.background = element_rect(fill = "grey95", color = NA),
```

```r
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
    ))
ggsave("fig_3c_shannon_qq_plots.png", plot = qq_plot_shannon, width = 10, height = 10, dpi = 300)

# Interaction Plot
plot_data_shannon <- table_shannon %>%
  group_by(City, Cutotype) %>%
  summarise(
    mean_value = mean(value),
    se_value = sd(value) / sqrt(n()),
    .groups = "drop"
  )
(interaction_plot_shannon <- ggplot(
  plot_data_shannon,
  aes(x = City, y = mean_value, group = Cutotype, color = Cutotype)
) +
    geom_point(size = 5) +
    geom_line(linewidth = 1.5) +
    geom_errorbar(
      aes(ymin = mean_value - se_value, ymax = mean_value + se_value),
      width = 0.15,
      linewidth = 1
    ) +
    scale_color_manual(values = c("#71b7ed", "#f57c6e")) +
    labs(
      title = "Interaction Plot: City × Cutotype on Shannon",
      x = "City",
      y = "Mean Shannon ± SE",
      color = "Cutotype"
    ) +
    theme_minimal() +
    theme(
      panel.grid.major.y = element_line(color = "grey95", linewidth = 1),
      panel.grid.minor = element_blank(),
      panel.grid.major.x = element_blank(),
      plot.margin = margin(15, 15, 15, 15),
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
```

```
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 1),
      legend.position = "top",
      strip.background = element_rect(fill = "grey95", color = NA),
      strip.text = element_text(face = "bold", size = 12),
      plot.subtitle = element_text(face = "bold", hjust = 1)
    ))
ggsave("fig_3c_shannon_interaction_plot.png", interaction_plot_shannon, width = 10, height = 10, dpi = 300)
```

4.) Beta-diversity analysis

```r
#####################################################################
# Preparation
#####################################################################

# Load packages (Install them first if have not)
library(readr)
library(dplyr)
library(reshape2)
library(vegan)
library(ggplot2)

# Set seed for random processes
set.seed(1)

#####################################################################
# 4.pre. Distance Matrices
#####################################################################

# Read the previously saved data
taxa_wide <- read_tsv("OTU_count_wide.tsv") %>% as.data.frame()
rownames(taxa_wide) <- taxa_wide$Sample
taxa_wide$Sample <- NULL

# Calculate the distance matrix
# Bray-Curtis dissimilarity
dm_bray <- vegdist(taxa_wide, method="bray") %>%  as.matrix()
# Jaccard distance
dm_jaccard <- vegdist(taxa_wide, method="jaccard", binary =
TRUE) %>%  as.matrix()

#####################################################################
# 4.a. PCoA
#####################################################################

# Read meta information
Meta <- read_tsv("metadata.txt")

# PCoA - bray
# Analysis
PCoA_bray <- cmdscale(dm_bray, k = 2, eig = TRUE)
```

```r
DF_bray <- data.frame(Sample = row.names(PCoA_bray$points), PCoA1 = PCoA_bray$points[,1], PCoA2 = PCoA_bray$points[,2], row.names = NULL)
Eigenvalues_bray <- eigenvals(PCoA_bray)
VarianceExplained_bray <- Eigenvalues_bray /sum(Eigenvalues_bray)
VarianceExplained1_bray <- 100 * signif(VarianceExplained_bray[1], 2) # unit: %
VarianceExplained2_bray <- 100 * signif(VarianceExplained_bray[2], 2) # unit: %
PCoA_bray <- merge(DF_bray, Meta, by = "Sample", all.x = TRUE)
# Visualize
PCoA_bray$City <- factor(PCoA_bray$City, levels = c("Baoding", "Dalian" ))
PCoA_bray$Cutotype <- factor(PCoA_bray$Cutotype, levels = c("Cutotype 1", "Cutotype 2"))
(pcoa_bray_plot <- ggplot(PCoA_bray, aes(x = PCoA1, y = PCoA2, color=Cutotype)) +
    geom_point(size=4, alpha = 0.9, aes(shape=City)) +
    stat_ellipse(
      geom = "polygon",
      alpha = 0.1,
      linewidth = 1,
      aes(fill = Cutotype, color = Cutotype)
    ) +
    xlab(xlab) +
    ylab(ylab) +
    labs(
      title = "PCoA of Bray-Curtis Dissimilarity",
      x = paste0("PCoA1 (Variance explained: ", VarianceExplained1_bray,"%)"),
      y = paste0("PCoA2 (Variance explained: ", VarianceExplained2_bray,"%)")
    ) +
    scale_fill_manual(values = c("#71b7ed", "#f57c6e")) +
    scale_color_manual(values = c("#71b7ed", "#f57c6e")) +
    theme_bw() +
    theme(
      legend.position = "right",
      legend.title = element_text(face = "bold"),
      legend.text = element_text(face = "bold"),
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
```

```r
      plot.subtitle = element_text(face = "bold", hjust = 1)
    ))
ggsave("fig_4a_bray_pcoa.png", width = 10, height = 10, dpi = 300)

# PCoA - jaccard
# Analysis
PCoA_jaccard <- cmdscale(dm_jaccard, k = 2, eig = TRUE)
DF_jaccard <- data.frame(Sample = row.names(PCoA_jaccard$points), PCoA1 = PCoA_jaccard$points[,1], PCoA2 = PCoA_jaccard$points[,2], row.names = NULL)
Eigenvalues_jaccard <- eigenvals(PCoA_jaccard)
VarianceExplained_jaccard <- Eigenvalues_jaccard /sum(Eigenvalues_jaccard)
VarianceExplained1_jaccard <- 100 * signif(VarianceExplained_jaccard[1], 2) # unit: %
VarianceExplained2_jaccard <- 100 * signif(VarianceExplained_jaccard[2], 2) # unit: %
PCoA_jaccard <- merge(DF_jaccard, Meta, by = "Sample", all.x = TRUE)
# Visualize
PCoA_jaccard$City <- factor(PCoA_jaccard$City, levels = c("Baoding", "Dalian" ))
PCoA_jaccard$Cutotype <- factor(PCoA_jaccard$Cutotype, levels = c("Cutotype 1", "Cutotype 2"))
(pcoa_jaccard_plot <- ggplot(PCoA_jaccard, aes(x = PCoA1, y = PCoA2, color=Cutotype)) +
    geom_point(size=4,alpha = 0.9, aes(shape=City)) +
    stat_ellipse(
      geom = "polygon",
      alpha = 0.1,
      linewidth = 1,
      aes(fill = Cutotype, color = Cutotype)
      ) +
    xlab(xlab) +
    ylab(ylab) +
    labs(
      title = "PCoA of Binary Jaccard Distance",
      x = paste0("PCoA1 (Variance explained: ", VarianceExplained1_jaccard,"%)"),
      y = paste0("PCoA2 (Variance explained: ", VarianceExplained2_jaccard,"%)")
    ) +
    scale_fill_manual(values = c("#71b7ed", "#f57c6e")) +
```

```r
    scale_color_manual(values = c("#71b7ed", "#f57c6e")) +
    theme_bw() +
    theme(
      legend.position = "right",
      legend.title = element_text(face = "bold"),
      legend.text = element_text(face = "bold"),
      plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
      axis.title = element_text(face = "bold", size = 12),
      axis.text = element_text(face = "bold"),
      plot.subtitle = element_text(face = "bold", hjust = 1)
    ))
ggsave("fig_4a_jaccard_pcoa.png", width = 10, height = 10, dpi = 300)

######################################################################
# 4.b. PERMANOVA
######################################################################

# Permanova - bray
# use `by="margin"` to prevent the influence of order
(adonis_bray <- adonis2(dm_bray ~ City + Cutotype + Age, data = Meta, by = "margin"))
# Permanova - jaccard
# use `by="margin"` to prevent the influence of order
(adonis_jaccard <- adonis2(dm_jaccard ~ City + Cutotype + Age, data = Meta, by = "margin"))
```